

Upasana Sridhar

Ph.D. Candidate, Electrical and Computer Engineering, Carnegie Mellon University

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Bio. My research asks how programming abstractions should evolve as software development becomes increasingly collaborative and agentic. I study how expert intent can be made explicit, reusable, and checkable across the layers of the computing stack.

In high-performance computing, I have explored this through abstractions and templates that translate high-level computational intent into implementations specialized to hardware constraints. More recently, I have become interested in how these same ideas change when programmers increasingly delegate implementation work to AI systems: what information must be expressed, what decisions should remain visible, and how generated implementations can be evaluated efficiently.

My broader goal is to develop programming models that help humans and AI systems collaborate while preserving the reasoning needed to understand, verify, and improve the resulting software.

Education

Carnegie Mellon University

Aug. 2019 – Present

Ph.D. Electrical and Computer Engineering, expected December 2026

M.S. Electrical and Computer Engineering, December 2026

QPA: 3.66

PES University, Bengaluru, India

Aug. 2015 – May 2019

B.Tech. Computer Science, 2019

GPA: 9.08/10.00

Honors and Awards

- **International HPC Summer School Selection** 2026
Selected as one of approximately 100 early-career researchers worldwide for an intensive international program in high-performance and scientific computing, with opportunities to engage with a global community of computational scientists.
- **IPDPS Ph.D. Forum Selection** 2025
One of 26 students selected to present dissertation research at the IEEE International Parallel and Distributed Processing Symposium Ph.D. Forum.
- **Proposal accepted to AMD Pervasive AI Contest** 2024
Received an AMD Ryzen Mini-PC to study the performance tradeoffs of the AIE accelerator.
- **CMU ECE Commitment to DEI Award** 2024
Recognized contributions to inclusion, belonging, and community-building within the Electrical and Computer Engineering department.
- **CMU CIT Dean's Ph.D. Fellowship Award** 2019
Fellowship covering tuition and stipend for the first year of Ph.D. study.

Publications

Peer-Reviewed Journal Articles

- **Upasana Sridhar**, Nikolai Tukanov, Elliott Binder, Tze Meng Low, Scott McMillan, and Martin D. Schatz. “SMaLL: Software for Rapidly Instantiating Machine Learning Libraries.” *ACM Transactions on Embedded Computing Systems (TECS)*, 2024.
- Sergio Barrachina, Adrián Castelló, Manuel F. Dolz, Tze Meng Low, Héctor Martínez, Enrique S. Quintana-Ortí, **Upasana Sridhar**, and Andrés E. Tomás. “Reformulating the Direct Convolution for High-Performance Deep Learning Inference on ARM Processors.” *Journal of Systems Architecture (JSA)*, 2023.

Peer-Reviewed Conference and Workshop Papers

- **Upasana Sridhar**, Elliott Binder, and Tze Meng Low. “Gen-AI in a Bottle: Experiments with LLMs to Generate HPC Kernels.” *IEEE IPDPS Workshops (IPDPSW)*, 2025.
- Yuttapichai Kerdcharoen, **Upasana Sridhar**, Orathai Sangpetch, and Tze Meng Low. “Linear Algebra Approach for Directed Triad Counting and Enumeration.” *SC24 Workshops*, pp. 718–726, 2024.
- Yuttapichai Kerdcharoen, **Upasana Sridhar**, and Tze Meng Low. “Exploiting Fusion Opportunities in Linear Algebraic Graph Query Engines.” *IEEE High Performance Extreme Computing Conference (HPEC)*, 2023.
- Tze Meng Low, Yuejie Chi, James C. Hoe, Swarun Kumar, Akarsh Prabhakara, Laixi Shi, **Upasana Sridhar**, Nicolai Tukanov, Chengyue Wang, and Yuchen Wu. “Zoom Out: Abstractions for Efficient Radar Algorithms on COTS Architectures.” *IEEE International Symposium on Phased Array Systems & Technology (PAST)*, 2022.
- **Upasana Sridhar**, Tze Meng Low, and Martin D. Schatz. “Fusing Non Element-wise Layers in DNNs.” *IEEE High Performance Extreme Computing Conference (HPEC)*, 2021.
- Ariful Azad, Mohsen Mahmoudi Aznaveh, Scott Beamer, Maia P. Blanco, Jinhao Chen, Luke D’Alessandro, Roshan Dathathri, Tim Davis, Kevin Deweese, Jesun Firoz, **Upasana Sridhar**, Tze Meng Low, and Martin D. Schatz. “Evaluation of Graph Analytics Frameworks Using the GAP Benchmark Suite.” *IEEE International Symposium on Workload Characterization (IISWC)*, 2020.
- Daniele G. Spampinato, **Upasana Sridhar**, and Tze Meng Low. “Linear Algebraic Depth-First Search.” *ACM SIGPLAN ARRAY Workshop*, pp. 93–104, 2019.
- **Upasana Sridhar**, Maia P. Blanco, Rahul Mayuranath, Daniele G. Spampinato, Tze Meng Low, and Scott McMillan. “Delta-Stepping SSSP: From Vertices and Edges to GraphBLAS Implementations.” *IEEE IPDPS Workshops*, 2019.
- Tze Meng Low, Daniele G. Spampinato, Anurag Kutuluru, **Upasana Sridhar**, Doru Thom Popovici, Franz Franchetti, and Scott McMillan. “Linear Algebraic Formulation of Edge-Centric k-Truss Algorithms with Adjacency Matrices.” *IEEE High Performance Extreme Computing Conference (HPEC)*, 2018.

Technical Reports, Briefs, and Other Publications

- Scott McMillan, Jay Palat, Oren Wright, **Upasana Sridhar**, Tze Meng Low, Nicolai Tukanov, Pankti Shah, and Navya Chandra. “PHITE: Portable High-Performance Inference at the Tactical Edge.” 2022.

- Scott McMillan, Tze Meng Low, Oren Wright, **Upasana Sridhar**, and Nicolai Tukanov. “Member Brief: SEI, AI Division, Advanced Computing Lab.” 2022.
- Scott McMillan, Elliott Binder, Mark Blanco, Paul Brower, Franz Franchetti, James C. Hoe, Tze M. Low, Peter Oostema, Fazle Sadi, Daniele Spampinato, and **Upasana Sridhar**. “Spiral/AIML: Frontiers of Graph Processing in Linear Algebra.” 2022.

Research Experience

Carnegie Mellon University — Graduate Research Assistant Aug. 2019 – Present

Advisor: Prof. Tze Meng Low

Thesis Title: Designing Extensible and Portable Performance Libraries for CNNs

Reduced the specialization burden of implementing a CNN library by reusing performance design across operators and hardware; supports 130 ONNX operators with 36 specialized microkernels organized into two dataflow patterns.

Microsoft Research — Research Intern

May 2023 – Aug. 2023

Redmond, WA

Manager: Srikanth Bharadwaj

Developed an analytical model for mixed-precision foundation-model inference to evaluate accuracy, latency, and throughput tradeoffs. Demonstrated that up to 20% of model weights could remain in high precision without degrading inference performance.

Microsoft Research — Research Intern

May 2021 – Aug. 2021

Managers: Patrice Godefroid and Marina Polishchuk

Worked on RESTler, a stateful REST API fuzzer, using production API traffic to improve request generation. Applied anomaly detection to identify fields causing request failures and developed a model to retry failed requests using predicted field values.

Carnegie Mellon University — Research Staff

Jan. 2019 – May 2019

Advisor: Prof. Tze Meng Low

Developed a code-generation framework for outer-product-like operations across applications including clustering and computational genetics. Abstracted their shared pairwise-operation-and-reduction structure to automatically generate high-performance x86 assembly microkernels.

Carnegie Mellon University — Research Intern

Jun. 2018 – Aug. 2018

Advisor: Prof. Franz Franchetti

Studied formal methods for deriving provably correct graph algorithms and applied these techniques to instantiate high-performance implementations of triangle counting on GPUs and k-truss on CPUs.

Invited Talks

- **SMaLL: Portable and Rapidly Instantiable Machine Learning Libraries**

- PES University – Nov. 2023
- Indian Institute of Science (IISc) – Nov. 2023
- BLIS Retreat – 2023
- Schlumberger (SLB) – Aug. 2025

- **Composable Abstractions for Layer Fusion: Convolution, Pooling, and Beyond**

- BLIS Retreat – 2024

- **Stories from the Field: Designing Production AI Systems**

– PES University – Mar. 2023

- **AI for Bottle Screening**

– CMKL University, Thailand – Jul. 2021

Conference Presentations and Posters

- **Gen-AI in a Bottle: Experiments with LLMs to Generate HPC Kernels**

Presentation, International Workshop on Automatic Performance Tuning (IWAPT), 2025.

- **Beyond Elementwise Fusion in CNNs**

Poster presentation, IEEE High Performance Extreme Computing Conference (HPEC), 2021.

- **Designing Reusable Performance Libraries for Edge ML**

Poster presentation, International HPC Summer School (IHPCSS), 2026.

- **A Unified Abstraction for Efficient and Portable CNN Libraries on Edge-Devices**

Poster presentation, IPDPS Ph.D. Forum, 2025.

Teaching and Mentoring

Courses Taught / Assisted

- **How to Write Fast Code II**

– **Guest Instructor** – Spring 2024, Spring 2026

Lectures on GPU Architecture and Programming; Task-based Parallelism in OpenMP.

– **Project Mentor / Subject-Matter Expert** – Spring 2023, Spring 2024

Provided questions and feedback during final presentations of student-led projects.

- **Computational Problem Solving for Engineers**

– **Teaching Assistant** — Spring 2025

Designed recitations and assignment questions, ran office hours and graded assignments.

- **How to Write Fast Code I**

– **Guest Instructor** — Fall 2022, Fall 2023

Delivered lectures introducing SIMD instructions, designing kernels for Convolutions.

– **Teaching Assistant** — Fall 2020, Fall 2021

Supervised 9 teams of 3-4 students working on the open-ended class projects. Ran office hours, graded homeworks, managed a hybrid classroom.

– **Project Mentor / Subject-Matter Expert** — Fall 2022, Fall 2023

Provided feedback to project teams on their designs and on their final project presentations.

- **Advanced Algorithms**

– **Teaching Assistant** — Fall 2018

– Project-based course

- **Nature of Code**

– **Instructor** — Jan.–Mar. 2018

– Student-taught course, PES University

Teaching Training

- Completed *Teaching As Research (TAR) Institute*, CMU Eberly Center, 2022. A 3-day program focused on developing skills to use data to inform teaching. Covered data sources, study design, and incorporating feedback and presenting student outcomes.
- Completed *Evidence-Based Teaching in STEM*, CMU, Spring 2024. A semester-long course designed to prepare graduate students in STEM disciplines to: (1) teach effectively and efficiently as future faculty members; (2) critically read and apply peer-reviewed, STEM-based educational research; and (3) adapt approaches from the Scholarship of Teaching and Learning (SoTL) to formatively assess student learning and iteratively improve teaching and course design. *Course Developed: How to Think in Code - Introduction to CS for non-majors*
- Completed *Future Faculty Program*, CMU Eberly Center, Spring 2026.
A program to help graduate students and postdocs develop teaching skills and document their teaching-related professional development activities in preparation for a faculty career.

Service

Professional Service

- **Workshop Co-Organizer** 2nd LLM4HPC Workshop, ISC High Performance, 2026.
- Program Committee Member, 1st International Workshop on Agentic AI for HPC, SC 2026
- Program Committee Member, 13th Workshop on Accelerator Programming and Directives, SC 2026.
- Program Committee Member, 1st Workshop on LLMs for HPC, SC Asia 2026.
- Program Committee Member, AI Track, International Conference on Parallel Processing (ICPP) 2025.

Departmental and University Service

- **Co-Chair**, Student Roundtable for Faculty Recruitment Interviews, CMU ECE, Spring 2023 and 2024.
- **ECE Open House Student Representative**, Carnegie Mellon University, 2020-Present.
- **DEI Student Advisory Committee**, CMU College of Engineering, November 2021 - May 2025.

Outreach and Broadening Participation

- **Pittsburgh Women in Mathematics and Computing Conference**: Planning & Publicity Chair, one-day workshop for undergraduates in the Pittsburgh area, February 2023.
- **Get Performance With Me: Pipelining in the Real World**
Speaker, GAINS Conference for high-school gender minorities in STEM.
- **STEM Outreach, CMU College of Engineering**, Spring 2021–Present. In-class BBC Microbit programming workshops and after-school STEM experiences.
- **Skype-a-Scientist**, Spring 2023. Introductory computer graphics session for middle school students.
- **Read-a-Story (India)**: Volunteer at a literacy and reading comprehension program.

Skills

Languages: C, C++, Python, x86 Assembly, ARM Assembly, SystemVerilog, Matlab, Lean4

Platforms & Frameworks: CUDA, Triton, AMD AIE/NPU, OpenMP, MPI, Slurm, PyTorch, TensorFlow, ONNX, TVM, GraphBLAS

Tools: NVIDIA Nsight/nvprof, Linux, perf, vtune, gem5, Git, CMake

Research Expertise: Reusable ML Library Design, Analytical Performance Modeling, Heterogeneous Computing, Software-Hardware Codesign, Hardware-Aware Applied ML, Distributed Computing, GPU Programming, Formal Methods for High-Performance Code